Hyun Joon Cha

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Research Interest

Real-world autonomous systems that bridge human-like reasoning with physical action.

Specific Topics: Reinforcement Learning, Data Infrastructure for AI, Tactile Sensing, Human Cognition, Dexterous Manipulation

Education & Military Service

Seoul National University Mar. 2025 – Feb. 2027(Planned)

M.S. in Mechanical Engineering

Advisor: Suk Won Cha, Stanford Ph.D., Director of SNU Engineering Research Institute, Director of SNU Next-Generation Automotive Research Center

Representative Coursework: Control System, Prompt Engineering, Natural Language Processing with Deep Learning, Knowledge and Database System

Korea University Mar. 2019 – Feb. 2025

B.S. in Chemical and Biological Engineering

Representative Coursework: Numerical Analysis, Process Control, Physical Chemistry, Biochemistry, Polymer Properties, Heat and Mass Transfer, Battery Dynamics

ROK Air Force - CBRN (Chemical Biological Radiological Nuclear) Defense Unit Oct. 2021 – Jul. 2023

Training in CBRN hazard detection, collection, and decontamination protocols

Research Experience

Stanford HAI Center, Stanford University
Visiting Student Researcher May. 2026 – Nov. 2026

- Designed LLM-based reward designing framework for FCHEV environments

Renewable Energy Conversion Lab, Seoul National University
Graduate Research Assistant Mar. 2025 – Present

- Designed Reinforcement Learning-based fuel cell power controller for Fuel Cell Hybrid Electric Vehicles to maximize fuel economy while minimizing fuel cell degradation.
- Designed a defect battery detection algorithm by analyzing large-scale electrochemical parameter data from battery cell production lines.

Undergraduate Research Assistant Jun. 2024 – Feb. 2025

- Implemented optimal vehicle trajectory tracking control using Model Predictive Control (MPC) and Pontryagin's Minimum Principle (PMP) to minimize tracking error.

System Biological Engineering Lab, Korea University
Undergraduate Research Assistant Nov. 2023 – May. 2024

- Increased antibiotic production efficiency in *Actinobacteria* by utilizing genomic recombination techniques.

Awards

Korea University Chemical Engineering Capstone Design - Top Prize Dec. 2024

- Optimized sensor type and deployment strategies in underground garages for immediate thermal runaway detection.

Conference Publication

Hyun Joon Cha, Donghwan Sung, Hyunjun Yang, Gwangeon Lee, Suk Won Cha*, "Reinforcement Learning-Based Energy Management System for Fuel Cell Hybrid Electric Vehicles Considering Fuel Cell Degradation", 2025 Korean Society of Automotive Engineers (KSAE) Autumn Conference

Skills

Familiar Software Tools: Python (Pytorch, Tensorflow), C/C++, MATLAB (Simulink, Simscape), PostgreSQL, Linux